

GOESGEN, Switzerland

WMO: 066260

Lat: 47.3631N

Lon: 7.9736E

Elev: 380

StdP: 96.84

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.2	(c) -5.9	(d) -12.0	(e) 1.4	(f) -7.0	(g) -8.9	(h) 1.8	(i) -4.4	(j) 11.8	(k) 7.2	(l) 10.6	(m) 7.6	(n) 2.3	(o) 50	(p) 0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	31.6	20.2	29.5	19.7	27.5	18.9	21.3	29.0	20.4	27.5	19.6	25.9	2.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.4	23.8	18.1	13.7	22.4	17.4	13.1	21.6	63.5	29.0	60.3	27.7	57.4	26.0	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	6.5	5.3	-11.3	34.6	3.0	1.4	-13.5	35.6	-15.3	36.5	-17.0	37.3	-19.2	38.3		
			-11.8	23.3	3.1	0.8	-14.1	23.9	-15.9	24.4	-17.6	24.9	-19.9	25.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	1.3	1.8	6.1	10.4	13.9	18.2	20.2	19.6	15.4	10.9	5.8	2.1		
	DBStd	7.56	3.66	4.08	3.41	3.45	3.52	3.33	3.08	3.13	3.03	3.58	3.48	3.50		
	HDD10.0	1078	269	230	126	37	7	0	0	0	1	34	130	245		
	HDD18.3	3038	527	463	379	237	142	43	17	23	96	231	376	503		
	CDD10.0	1273	1	1	4	49	128	244	317	298	164	62	5	0		
	CDD18.3	189	0	0	0	0	6	38	75	62	8	0	0	0		
	CDH23.3	1830	0	0	0	13	85	403	719	534	74	1	0	0		
	CDH26.7	601	0	0	0	1	13	131	264	181	11	0	0	0		
(14) Wind	WSAvg	1.7	2.2	1.9	2.1	1.6	1.7	1.5	1.4	1.2	1.3	1.3	1.7	2.1		
(15) Precipitation	PrecAvg	1093	73	69	70	82	106	111	117	110	88	87	88	93		
	PrecMax	1383	187	168	248	179	183	230	204	213	157	163	211	199		
	PrecMin	752	16	23	24	9	34	26	37	17	42	11	4	0		
	PrecStd	163	44	37	49	45	44	47	37	46	36	41	50	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	15.4	19.6	25.4	28.6	33.3	34.7	33.4	28.2	22.2	16.9	11.6		
		MCWB	8.8	8.9	11.0	14.2	18.5	21.6	21.0	20.7	18.7	15.3	12.1	9.2		
	2%	DB	9.6	11.7	17.6	22.6	25.5	30.3	32.0	30.8	25.8	20.0	14.2	10.3		
		MCWB	7.5	7.2	10.0	12.9	16.4	20.1	19.7	20.6	18.2	14.6	11.1	8.2		
	5%	DB	7.8	9.3	15.3	20.7	23.7	27.6	29.7	28.7	23.4	17.8	12.4	8.7		
		MCWB	5.9	6.3	9.1	12.6	15.4	19.0	19.5	19.9	16.8	14.0	10.0	6.9		
	10%	DB	6.4	7.1	12.7	18.3	21.3	25.4	27.4	26.3	21.3	16.0	10.6	7.2		
		MCWB	4.9	4.8	7.9	11.6	14.5	18.1	18.6	19.1	16.1	13.2	8.8	5.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	9.6	11.8	15.0	19.1	22.9	22.3	22.2	19.7	16.3	12.7	9.5		
		MCDB	11.3	13.6	17.8	23.1	26.5	31.4	31.1	29.5	25.5	20.3	15.7	11.1		
	2%	WB	7.7	7.6	10.5	13.9	17.6	21.0	21.0	21.2	18.6	15.3	11.4	8.4		
		MCDB	9.3	11.3	16.2	20.7	23.7	28.4	29.1	28.5	24.1	18.6	14.0	10.0		
	5%	WB	6.2	6.3	9.4	12.9	16.2	19.7	20.1	20.4	17.7	14.5	10.1	7.2		
		MCDB	7.7	9.1	14.2	19.4	21.6	26.1	27.4	27.2	22.1	17.0	12.2	8.6		
	10%	WB	5.0	4.9	8.3	11.9	15.1	18.6	19.3	19.6	16.7	13.6	9.0	5.8		
		MCDB	6.3	6.9	12.3	17.7	20.2	24.3	25.7	25.3	20.2	15.6	10.2	7.0		
(35) Mean Daily Temperature Range	MDBR		4.7	6.9	10.0	11.5	10.4	10.9	11.5	11.0	10.0	7.9	5.5	4.8		
	5% DB	MCDBR	7.1	11.6	15.7	17.0	15.2	15.5	16.3	15.1	13.4	10.8	8.6	6.4		
		MCWBR	5.2	7.5	8.8	8.5	7.0	6.6	5.8	5.9	6.3	6.0	5.7	4.8		
	5% WB	MCDBR	6.9	10.9	13.9	15.6	13.0	14.0	14.2	13.6	11.6	9.8	8.0	6.0		
		MCWBR	5.2	7.1	8.0	8.1	6.3	6.2	5.5	5.7	5.7	5.7	5.5	4.7		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.326	0.362	0.395	0.407	0.411	0.412	0.404	0.381	0.358	0.323	0.297		
	taud		2.440	2.367	2.316	2.257	2.268	2.300	2.291	2.327	2.374	2.421	2.464	2.448		
	Ebn at Noon		788	835	854	857	858	855	847	839	825	787	750	745		
	Edh at Noon		67	88	108	126	130	127	126	117	102	83	65	59		
(44) All-Sky Solar Radiation	RadAvg		1.16	1.99	3.21	4.34	5.01	5.84	5.63	4.90	3.71	2.34	1.28	0.94		
	RadStd		0.14	0.29	0.47	0.64	0.49	0.54	0.51	0.48	0.36	0.24	0.16	0.16		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (117 neighbors)	N/A	N/A	N/A	+0.73	+0.42	+0.32	N/A	N/A	+56	+14

Nomenclature: See separate page